PART 07 — REST & SUBLEDGER API SPECIFICATION

API Base URL: `http://localhost:4000/api/v1` (Production: `https://api.blynt.co.uk/api/v1`)

Data Exchange Format: JSON (`application/json; charset=utf-8`)

Security Standard: Bearer Token via HTTP Header `Authorization: Bearer <token>`

Correlation Tracking: Unique `x-request-id` header generated on every incoming request

1. Global Architectural Standards

1.1 Standard Domain Error Response Envelope

Every non-2xx error returns a standardized JSON envelope matching `DomainException`:

`json

{

"statusCode": 400,

"errorCode": "ERR_JOURNAL_UNBALANCED",

"message": "Total debits (£1,000.0000) must equal total credits (£950.0000). Imbalance: £50.0000",

"details": {

"totalDebits": "1000.0000",

"totalCredits": "950.0000",

"difference": "50.0000"

},

"timestamp": "2026-10-11T23:30:00.000Z",

"path": "/api/v1/journals/clx49817203/validate",

"requestId": "req_847291048291"

}

1.2 Domain Error Catalog (Authoritative Codes)

Error Code	HTTP Status	Meaning
`ERR_AUTH_INVALID_CREDENTIALS`	401	Email or password incorrect.
`ERR_AUTH_SESSION_EXPIRED`	401	Bearer token revoked or expired.
`ERR_AUTH_FORBIDDEN`	403	User role lacks permission for this endpoint.

`ERR_JOURNAL_UNBALANCED`	400	Debits do not equal credits down to 4 decimal places.
`ERR_JOURNAL_MINIMUM_LINES`	400	Journal must have at least 2 distinct lines.
`ERR_CONTROL_ACCOUNT_DIRECT_POSTING`	400	Manual entry to protected control accounts is forbidden.
`ERR_PERIOD_NOT_FOUND`	404	Transaction date does not resolve to an accounting period.
`ERR_PERIOD_HARD_LOCKED`	403	Period has been permanently locked for statutory filing.
`ERR_ORGANIZATION_SUSPENDED`	403	Target company workspace is suspended or archived.
`ERR_TENANT_VIOLATION`	403	Attempted cross-tenant reference detected.

2. API Endpoints by Domain

2.1 Authentication Endpoints (`/auth`)

- `POST /auth/register`: Create user account, hashed with `scrypt`, and provisions demo organisation.
- `POST /auth/login`: Validates credentials with constant-time equality check; returns session token.
- *Request Body:* `{"email": "accountant@company.com", "password": "Password1234!"}`
- *Response 200:* `{"accessToken": "sess_tok_92819...", "user": {...}, "organizations": [...]}`
- `POST /auth/logout`: Revokes active session token.
- `GET /auth/sessions`: Returns list of active user sessions with IP, user-agent, and login timestamps.

2.2 Organizations Endpoints (`/organizations`)

- `GET /organizations`: Lists all workspaces the current user has access to.
- `POST /organizations`: Provisions a new company workspace with seeded UK Chart of Accounts.
- `GET /organizations/:id`: Retrieves legal name, company number, and VAT settings.
- `PATCH /organizations/:id`: Updates company metadata.

2.3 Chart of Accounts Endpoints (`/accounts`)

- `GET /accounts`: Retrieves nominal ledger accounts (supports filtering by `type=ASSET` or `status=ACTIVE`).
- `POST /accounts`: Adds a custom nominal account.
- *Request Body:* `{"code": "5050", "name": "Cloud Hosting Fees", "type": "EXPENSE", "subtype": "OPERATING_EXPENSE"}`
- `POST /accounts/:id/archive`: Archives an account.

2.4 Accounting Periods Endpoints (`/financial-years`, `/accounting-periods`)

- `GET /financial-years`: Lists financial years and calendar dates.
- `POST /financial-years`: Automatically generates a financial year and 12 monthly periods.
- `PATCH /accounting-periods/:id/lock`: Updates lock state (`OPEN`, `SOFT_LOCKED`, `HARD_LOCKED`).

2.5 Journals Endpoints (`/journals`)

- `GET /journals`: Returns paginated journal entries with status filtering.
- `POST /journals`: Creates a draft journal entry.
- *Request Body:*

`json

{

"entryDate": "2026-10-15",

"description": "Monthly Software Depreciation",

"lines": [

{ "accountId": "acc_6500", "debit": "500.0000", "credit": "0.0000", "description": "Depreciation expense" },

{ "accountId": "acc_1810", "debit": "0.0000", "credit": "500.0000", "description": "Accumulated depreciation" }

]

}

- `POST /journals/:id/validate`: Runs the 15-rule validation engine and transitions status to `VALIDATED`.
- `POST /journals/:id/reverse`: Generates a GAAP counter-balancing reversal journal.

2.6 Contacts & Subledger Endpoints (`/contacts`)

- `GET /contacts`: Lists customers and suppliers with real-time derived balance metrics.
- `POST /contacts`: Creates customer or supplier record with UK VAT number and payment terms.
- `GET /contacts/:id/statement`: Returns chronological statement with running line balances.

2.7 Sales Invoicing Endpoints (`/invoices`)

- `POST /invoices`: Creates draft sales invoice with item lines and UK VAT rate (20%, 5%, 0%).
- `POST /invoices/:id/post`: Approves invoice and auto-generates balanced double-entry GL journal.